Sudhanvi Vakul Vakulabharanam
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SUMMARY

ASIC / RTL / SoC Design Engineer with strong academic grounding and hands-on experience in digital design, computer architecture, microarchitecture, and system-level hardware development. Experienced in Verilog/SystemVerilog RTL design, CDC/RDC-safe methodologies, reset and clocking strategies, and AMBA-based SoC interconnects (AXI/APB). Proven ability to work across hardware–software boundaries through FPGA-based SoC prototyping, performance and latency profiling on GPUs and SoCs, and RTL quality checks. Strong background in Python-based automation, hardware acceleration, and ML-aware system optimization. Demonstrated leadership through technical project ownership, cross-functional coordination, mentoring, and delivery of reusable system-level IP and documentation.

EDUCATION

Illinois Institute of Technology, Chicago, IL, USA Master of Science – Electrical and Computer Engineering | GPA: 3.66 August 2023 – May 2025
Relevant Coursework: RF Integrated Circuits, VLSI Design Circuits, CMOS Circuits, Computer Architecture & Design, Hardware Acceleration Using ML, Computer Vision, Digital Signal & Image Processing, Hardware–Software Co-Design,
Bhoj Reddy Engineering College for Women, Hyderabad, Telangana, India Bachelor of Science – Electronics and Communication Engineering | GPA: 3.51 June 2016 – September 2020
Relevant Coursework: VLSI Design, Embedded Systems, Microprocessors and Microcontrollers, PCB Design, Analog & Digital Communications, Semiconductor Devices & Electronic Circuits, Electromagnetic Wave and Transmission Lines, Optical Communications, Digital Logic Design, GPS

TECHNICAL SKILLS

Digital Design & VLSI RTL Design: ASIC Design Flow, Microarchitecture Definition, FSM Design, SoC Architecture, AXI/APB Interconnects, CDC/RDC Analysis, Clock & Reset Strategies, Latency Analysis, RTL Quality Checks, STA/SDC, PLL Clocking, Synthesis & PnR, Timing Closure, PVT & Corner Analysis, Mixed-Signal Fundamentals, SRAM Design (6T/10T), Power/Area Optimization, Technology Nodes (7nm, 45nm)
Programming & Scripting Python: Verilog, SystemVerilog, VHDL, UVM, C/C++, Embedded C, MATLAB, TCL, Shell/Bash, Java, SQL, PyTorch, NumPy
EDA & Verification Tools: Xilinx Vivado/ISE, ModelSim, Cadence Innovus & Virtuoso, Synopsys VCS, ICC2, Design Compiler, PrimeTime, HSPICE, Finesim, Altium Designer, Vitis HLS
FPGA & Embedded Platforms: Kintex KC705, PYNQ-Z2, Spartan-7, MicroBlaze, RISC-V, Arduino, ESP32, 8051
Hardware Acceleration & ML Systems Quantization, Graph Optimization, FPGA Inference, Feature-Map Packing, Latency & Resource Profiling, Chipscope Debug, JTAG, PCB Design & Fabrication
Protocols & Debug: UART, SPI, I2C, Ethernet, PCIe, JTAG, Testbench Development, Functional Simulation, Waveform Debug, SVA, Coverage, Linting, Static Checks
Other Tools: Wireshark, TeraTerm, Oscilloscope, Spectrum Analyzer, Docker, Kubernetes, Kafka, Jira, Git, VS Code, Eclipse, Linux, Windows, macOS

WORK EXPERIENCE

Electronics Design Engineer (Part-Time) – Community Dreams Foundation (CDF) | September 2025 – Present
Programs: VLSIVault (Individual Project Owner), OpenHACCP (Team Project – 7–10 Members)
Skills & Tools: Verilog/SystemVerilog, RTL & SoC Microarchitecture, CDC/RDC Analysis, SystemVerilog Verification, Python, NLP Pipelines, Cloud Systems, Performance Monitoring, Technical Leadership
Technical Contributions:

- Architected and developed multiple RTL and SoC-level projects covering microarchitecture design, FSM-based control, clock/reset strategies, and AMBA-based interconnects (AXI/APB).
- Designed reusable Verilog/SystemVerilog RTL modules with well-defined interfaces, timing assumptions, and parameterization for scalability.
- Applied CDC- and RDC-safe design methodologies including synchronizers, handshake logic, and reset sequencing to ensure robustness across clock domains.
- Developed comprehensive SystemVerilog testbenches incorporating assertions (SVA), functional coverage, and directed/constrained-random stimulus.
- Performed RTL quality checks including latency analysis, critical-path identification, and waveform-level debugging using ModelSim and Vivado.
- Designed and owned NLP pipelines to ingest unstructured inputs such as recipes and process descriptions and convert them into structured, machine-readable representations.
- Implemented entity extraction and classification to identify ingredients, processing steps, hazards, and Critical Control Points (CCPs).
- Automated generation of fully formatted, standards-compliant HACCP documentation.

Other Contributions:

- Standardized coding guidelines, verification templates, and documentation practices to improve design quality and reusability across projects.
- Authored detailed technical documentation explaining architecture decisions, performance trade-offs, and verification strategies.
- Led and mentored a team of 7–10 engineers, conducting daily stand-ups, weekly planning, and design reviews.
- Defined milestones, assigned tasks, tracked progress, and ensured timely delivery of project artifacts.
- Acted as project owner, coordinating efforts across AI/ML engineers, backend developers, and food-safety domain experts.
- Led requirements definition, technical roadmap planning, and feature prioritization, Ensured alignment between regulatory requirements, technical implementation, and user needs.

Research Assistant (Part-Time) - Illinois Institute of Technology, Chicago, IL | September 2024 – February 2025

Primary Role: Machine Learning Researcher with Hardware Acceleration & Deployment Focus

Skills & Tools: Python, PyTorch, Transformer Architectures (MobileViT), GPU/CUDA, Conda, Quantization, Graph Optimization, FPGA Inference, PYNQ-Z2, GPU & SoC Profiling, RSCD 1M Dataset, ImageNet-1K

Technical Contributions:

- Researched and evaluated lightweight transformer-based perception models with emphasis on deployment feasibility for edge and embedded systems.
- Led experimentation with MobileViT transformer architectures, achieving 93% classification accuracy on the RSCD 1 Million dataset through systematic data-augmentation strategies.
- Designed and applied advanced data augmentation pipelines to improve generalization and robustness without increasing model complexity.
- Benchmarked model performance on large-scale datasets including RSCD 1M and ImageNet-1K, analyzing accuracy–latency trade-offs.
- Built an end-to-end FPGA inference pipeline on the PYNQ-Z2 platform, including test-vector generation, host–accelerator interfacing, and result validation.
- Profiled inference latency, throughput, and resource utilization on both GPUs and SoCs to identify performance bottlenecks.

- Debugged hardware–software mismatches using waveform tracing, intermediate tensor inspection, and cross-verification with software reference outputs.
- Applied post-training quantization and graph-level optimizations to reduce model size and inference cost while preserving accuracy.

Other Contributions:

- Collaborated with other research engineers and architects to deliver system-level IP and ensure hardware-software co-design alignment.
- Documented performance considerations and methodologies for knowledge sharing with cross-functional teams.

System Engineer (Full-Time) – Tata Consultancy Services, Hyderabad, India | August 2020 – July 2023

Skills & Tools: Workflow Automation, SQL, Systems Engineering, Cross-Team Coordination, Process Optimization, TOSCA, SAP

Technical Contributions:

- Led a high-complexity workflow modernization initiative spanning requirement analysis, design, implementation, UAT, and production rollout.
- Designed and implemented automated validation workflows using SQL-based rule engines to replace manual checks.
- Improved system reliability and consistency by reducing human error and standardizing validation logic across teams.

Other Contributions:

- Coordinated delivery across geographically distributed engineering teams, managing timelines, dependencies, and defect tracking.
- Maintained detailed technical and process documentation to support audits, onboarding, and long-term maintainability.
- Acted as a module lead, ensuring adherence to delivery schedules and quality benchmarks under time-critical constraints.

Project Apprentice (Full-Time) – DRDO, Hyderabad, India | January 2019 – April 2020

Project: UART to Ethernet Conversion Module

Skills & Tools: Verilog RTL, Vivado IP Integrator, Ethernet, UART, PCIe, KC705 FPGA, JTAG, Altium Designer

Technical Contributions:

- Designed a bidirectional UART-to-Ethernet bridge using Verilog RTL, implementing buffering, data framing, error detection, and flow control mechanisms.
- Integrated the design using Vivado IP Integrator and ensured protocol correctness under variable baud rates and traffic patterns.
- Verified functionality through simulation and on-hardware testing using TeraTerm for UART and Wireshark for Ethernet packet inspection.
- Performed static timing analysis and timing closure to ensure reliable operation across operating conditions.
- Programmed and debugged the KC705 FPGA via JTAG, monitoring real-time traffic and diagnosing corner-case failures.
- Designed and routed a custom PCIe-based PCB in Altium Designer, performing DRC, impedance checks, and signal-integrity considerations.
- Worked under the guidance of senior scientist Dr. Narasimhulu, gaining exposure to defense-oriented hardware development workflows.

Project Apprentice (Full-Time) – Bharat Dynamics Limited, Hyderabad, India | April 2018 – June 2018
Project: Through-Wall Object Detection Using RADAR
Skills & Tools: MATLAB, Radar Signal Processing, Modeling & Simulation
Technical Contributions:

- Developed MATLAB-based signal-processing models to analyze reflected radar signals through wall materials with known dielectric properties.
- Modeled attenuation, multipath propagation, scattering effects, and Doppler signatures.
- Distinguished moving human targets from static objects using frequency and phase-based analysis.

Other Contributions:

- Contributed to applied research by providing simulation results for team review and refinement.
- Documented modeling approaches to facilitate knowledge transfer within the project team.

PROJECT EXPERIENCE

FPGA-Based Systolic Array Accelerator for VGG11 (Hardware Acceleration Project)
Skills & Tools: C++, Vitis HLS, FPGA Architecture, Systolic Arrays, Fixed-Point Quantization, BRAM Optimization, PYNQ-Z2, PyTorch, Jupyter Notebook, Latency & Resource Profiling, modelsim, hspice

- Designed a parameterized systolic array architecture using C++ kernels in Vitis HLS, mapping convolution and matrix-multiplication operations onto processing elements (PEs) to maximize data reuse and parallelism.
- Defined microarchitectural parameters such as PE count, tile size, and dataflow strategy to balance throughput, resource utilization, and clock frequency.
- Implemented fixed-point quantization to reduce DSP utilization while maintaining acceptable inference accuracy.
- Architected a BRAM-partitioned on-chip memory hierarchy with double buffering to overlap computation with data movement.
- Optimized feature-map tiling and weight reuse to minimize external memory bandwidth requirements on the PYNQ-Z2 FPGA.
- Applied HLS directives such as loop pipelining, unrolling, and array partitioning to improve initiation interval (II) and throughput.
- Conducted design-space exploration using HLS C-simulation, co-simulation, and RTL simulation to evaluate performance trade-offs.
- Analyzed FPGA resource utilization (DSPs, LUTs, FFs, BRAM) and end-to-end latency across multiple configurations.
- Verified numerical correctness by comparing FPGA inference outputs against PyTorch reference models using Jupyter Notebook, achieving MSE ˜ 0.0039.
- Instrumented the design for latency and performance monitoring to identify bottlenecks and guide architectural refinements.

Image Classification Using YOLOv2 on ImageNet-1K
Skills & Tools: Python, YOLOv2, Deep Learning, Google Colab, Google Drive, ImageNet-1K, GPU-based Training, Model Evaluation

- Set up the training and evaluation pipeline using Google Colab, integrating Google Drive for efficient dataset storage and checkpoint management.
- Configured YOLOv2 architecture for ImageNet-1K classification tasks with darknet-19 backbone architecture, managing dataset preprocessing, batching, and label mapping.
- Executed training and inference using GPU acceleration available through Colab, monitoring training stability and convergence behavior.

- Analyzed classification accuracy and loss trends across training epochs.
- Studied the trade-offs between dataset scale, training time, and compute resource utilization.
- Gained practical exposure to cloud-based ML experimentation and large-scale dataset handling.

Low-Power 10T SRAM Using Sleep Transistors and Drain-Gating Techniques

Skills & Tools: Cadence Virtuoso, GPDK 45nm, HSPICE, Spectre, Low-Power VLSI Design, SRAM Stability Analysis

- Designed a custom 10T SRAM bitcell in Cadence Virtuoso incorporating Sleepy Keeper and Drain-Gating techniques to suppress leakage during idle modes.
- Carefully sized access, pull-up, and pull-down transistors to balance read/write margins and performance.
- Integrated sleep transistors without compromising data retention under low-voltage operation.
- Performed transient, DC, and leakage simulations using Spectre and HSPICE to evaluate read delay, write delay, and standby power.
- Analyzed Static Noise Margin (SNM) for read, write, and hold conditions across PVT corners.
- Benchmarked the proposed 10T cell against conventional 6T and baseline 10T SRAM designs, demonstrating reduced leakage power and improved read stability.
- Evaluated feasibility for SoC-level integration under voltage and temperature variations.

6T SRAM Cell Design Using ASAP 7nm Technology

Skills & Tools: Cadence Virtuoso, ASAP 7nm PDK, Calibre DRC/LVS, HSPICE, Parasitic Extraction, SNM Analysis

- Designed and laid out a complete 6T SRAM bitcell in Cadence Virtuoso following ASAP 7nm design rules.
- Completed full DRC and LVS verification using Calibre, ensuring layout correctness and schematic consistency.
- Performed parasitic extraction to capture realistic interconnect effects at nanoscale dimensions.
- Conducted transistor-level read/write simulations in HSPICE to analyze access time, leakage, and dynamic behavior.
- Performed multi-corner voltage and temperature sweeps to assess robustness under process variation.
- Generated butterfly curves to quantify read and write SNM, identifying stability degradation trends at low Vdd.
- Analyzed bitline discharge behavior, access transistor timing, and cell-level failure modes using waveform-level debugging.

Object Detection Using DINO-DETR

Skills & Tools: PyTorch, Transformer Architectures, Computer Vision, Model Optimization, Performance Evaluation

- Implemented and evaluated DAB-DETR, DN-DETR, and DINO-DETR techniques to address slow convergence and localization issues in standard DETR.
- Integrated dynamic anchor boxes and contrastive denoising with noisy queries to stabilize training.
- Applied dynamic query selection and Look-Forward Twice strategies to improve detection of small and overlapping objects.
- Trained and evaluated models on the PASCAL VOC dataset.
- Benchmarked performance using mAP, AR, and loss metrics.
- Achieved faster convergence, lower overall loss, and improved bounding-box alignment compared to baseline DETR.

Smartphone Controlled Arduino Bluetooth Robot Car

Skills & Tools: Arduino UNO, Embedded C/C++ (Arduino IDE), HC-06 Bluetooth Module, L298N Motor Driver, DC Motors (4WD), Proximity Sensor, Android Bluetooth RC Car App, PWM Motor Control, Serial Communication (UART)

- Designed and built a smartphone-controlled 4WD robot car using an Arduino UNO and HC-06 Bluetooth module for wireless command/control.

- Implemented an embedded motor-control stack using the L298N driver with PWM-based speed control and direction logic for differential steering.
- Developed Arduino firmware to parse Bluetooth serial commands (UART) from an Android RC car app and map them to motion primitives (forward/reverse/left/right/stop).
- Integrated a proximity sensor to enable real-time obstacle detection and automatic stop/avoid behavior to reduce collision risk.
- Performed end-to-end system integration and validation of Bluetooth connectivity range, command latency, and motor response under varying loads.
- Debugged control issues related to motor driver switching, power stability, and noisy sensor readings using serial logs and iterative tuning.
- Documented wiring, pin mapping, and calibration parameters (PWM thresholds, sensor trigger distance) to ensure repeatable bring-up and demos.

Extraction of Audio from Visual Information (Study Project)

Skills & Tools: Deep Learning, Computer Vision, Signal Processing, Audio Reconstruction, Python

- Studied the underlying research methodology proposed by MIT for recovering sound from high-speed visual input.
- Analyzed how subtle pixel-level vibrations in video sequences can be mapped to audio waveforms.
- Reproduced key components of the approach as a learning exercise, focusing on model structure, data representation, and reconstruction accuracy.
- Developed an understanding of multimodal learning and the relationship between vision and acoustics.
- Gained insight into challenges such as noise sensitivity, frame-rate dependency, and reconstruction fidelity.
- Strengthened ability to read, interpret, and replicate advanced research work.

System Exploration Project – Swish vs ReLU in Apache TVM

Skills & Tools: Apache TVM, Relay IR, AutoTVM, LLVM, Python, Compiler Optimization, Performance Benchmarking, CPU Profiling

- Studied the Apache TVM compilation stack including frontend ingestion, Relay Intermediate Representation (IR), Relay optimization passes, Tensor Expression (TE), Tensor IR (TIR), auto-scheduling via AutoTVM/Ansor, and LLVM-based code generation.
- Built a clear understanding of how high-level ML models are progressively lowered into hardware-optimized executables.
- Installed and configured Apache TVM (v0.21.dev0) on an Ubuntu 20.04 CPU-only environment using Conda and LLVM 10.0.0.
- Implemented two functionally equivalent two-layer MLP models in Relay: one using ReLU and the other using a custom Swish activation defined as x * sigmoid(x).
- Integrated the custom Swish operator using Relay's native operations and ensured correct lowering through the TVM compilation pipeline.
- Compiled both models for the LLVM CPU target and executed inference with randomized inputs.
- Measured CPU processing time, memory usage, and wall-clock latency using Python profiling utilities and TVM's built-in benchmarking framework.
- Observed that Swish incurs higher execution time and memory usage than ReLU due to the additional sigmoid computation, aligning with theoretical expectations.
- Demonstrated quantitative performance differences between simple and complex activation functions in a compiler-optimized environment.
- Highlighted the importance of activation-function selection in latency- and resource-constrained systems.

- Identified future opportunities for hardware–software co-design, including custom accelerator support for complex activations.

CERTIFICATIONS

- Fundamentals of System Verilog, System Verilog Projects, UVM Fundamentals from Namaste FPGA
- Design Certifications in CDC, STA, I2C, SPI, UART, AMBA AXI, AMBA APB in Verilog
- Won awards in many technical quizzes, circuit designing contests, seminars, debates